
The case for 4-bit precision: k-bit Inference Scaling Laws

Tim Dettmers¹ Luke Zettlemoyer¹

Abstract

Quantization methods reduce the number of bits required to represent each parameter in a model, trading accuracy for smaller memory footprints and inference latencies. However, the final model size depends on both the number of parameters of the original model and the rate of compression. For example, a 30B 8-bit model and a 60B 4-bit model have the same number of bits but may have very different zero-shot accuracies. In this work, we study this trade-off by developing inference scaling laws of zero-shot performance in Large Language Models (LLMs) to determine the bit-precision and model size that maximizes zero-shot performance. We run more than 35,000 experiments with 16-bit inputs and k-bit parameters to examine which zero-shot quantization methods improve scaling for 3 to 8-bit precision at scales of 19M to 176B parameters across the LLM families BLOOM, OPT, NeoX/Pythia, and GPT-2. We find that it is challenging to improve the bit-level scaling trade-off, with the only improvements being the use of a small block size – splitting the parameters into small independently quantized blocks – and the quantization data type being used (e.g., Int vs Float). Overall, our findings show that 4-bit precision is almost universally optimal for total model bits and zero-shot accuracy.

1. Introduction

Large Language Models (LLMs) are widely adopted for zero/few-shot inference (Zhang et al., 2022; Black et al., 2022; Zeng et al., 2022; Scao et al., 2022), but they can be challenging to use both due to their large memory footprints – up to 352 GB of GPU memory for 175B models – and high latency. However, both the memory and latency are

¹University of Washington
<dettmers@cs.washington.edu>.

Proceedings of the 40th International Conference on Machine Learning, Honolulu, Hawaii, USA. PMLR 202, 2023. Copyright 2023 by the author(s).

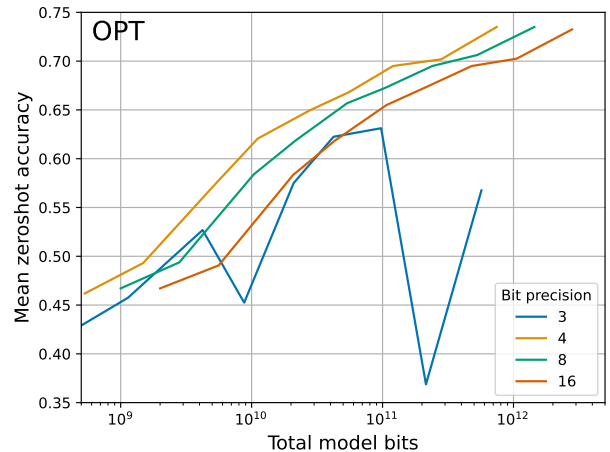


Figure 1. Bit-level scaling laws for mean zero-shot performance across four datasets for 125M to 176B parameter OPT models. Zero-shot performance increases steadily for fixed model bits as we reduce the quantization precision from 16 to 4 bits. At 3-bits, this relationship reverses, making 4-bit precision optimal.

primarily determined by the total number of bits in the parameters. Therefore, if we reduce the model bits through quantization, we can expect the latency of the model to reduce proportionally, potentially at the expense of end task accuracy (Frantar et al., 2022; Park et al., 2022; Yao et al., 2022).

Since we can quantize the parameters of a trained model to an arbitrary bit-precision, this raises the question of how many bits should be used to optimally trade off accuracy and total model bits, given our current base methods for model quantization. For example, **if we have a 60B LLM in 4-bit precision and a 30B LLM in 8-bit precision, which will achieve higher accuracy?** To study such trade-offs, it is helpful to take the perspective of scaling laws (Kaplan et al., 2020; Henighan et al., 2020), which evaluate the underlying trends of variables to make generalizations beyond individual data points.

In this paper, we study bit-level inference scaling laws for zero-shot quantization to determine the precision that maximizes zero-shot accuracy given a certain number of total bits in the model. **Our main finding is that 4-bit parameters yield optimal performance for a fixed number of model**